
Automated Multi-Agent Pipeline for Biomarker Discovery in Depression using Digital Wearable Data

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Abstract

We present an automated multi-agent data science pipeline for biomarker discovery in Depression. Applying the pipeline to a dataset of 7,497 observations and 159 features, we identified 44 statistically significant feature associations with the target variable (phq_score). Machine learning models for regression achieved best performance with Lasso ($R^2=0.244$, $RMSE=4.769$). The top biomarker candidates include financial_ordinal, age, unlocks_weekend_mean, unlocks_weekday_mean, sleep_woken_up_count_main_sleep_std. This automated approach demonstrates the potential of wearable and digital biomarkers for passive, scalable monitoring of Depression.

1 Introduction

Depression represents a significant public health challenge. Digital wearable devices offer a promising avenue for passive, continuous monitoring of behavioural and physiological biomarkers associated with Depression.

DWB Hourly (Digital Wellbeing Study) - Pre-aggregated wearable and smartphone behavioral data from 7,497 participants. Raw data was collected at hourly resolution and aggregated to one row per participant (cross-sectional). Features capture circadian rhythms, phone usage patterns, pre-sleep screen exposure, nocturnal behavior, activity, sleep architecture, and HRV-derived metrics. Target is PHQ-9 depression severity score. Only digital sensor-derived features should be used — self-report questionnaire scores (personality, sleep quality surveys, GAD, PSS) are excluded to avoid method variance.

In this study, we apply an automated multi-agent data science pipeline to systematically discover and validate digital biomarkers for Depression. The pipeline integrates data preprocessing, exploratory analysis, statistical hypothesis testing, and machine learning to identify candidate biomarkers from high-dimensional wearable sensor data.

2 Methods

2.1 Dataset Description

Table 1: Dataset summary.

Property	Value
Dataset	dwb_hourly
Disease / Condition	Depression
Task Type	Regression
Observations	7,497
Subjects	7,497
Features	159
Target Variable	phq_score

Feature categories:

- **Circadian Rhythm** (13 features): Cosinor-fitted circadian parameters for steps and phone unlocks
- **Screen And Phone** (25 features): Pre-sleep screen time, nocturnal phone use, unlock counts by hour
- **Sleep Objective** (30 features): Objectively measured sleep duration, efficiency, restlessness, architecture
- **Activity** (20 features): Step counts, activity intensity, sedentary behavior patterns
- **Hrv Cardiac** (20 features): Heart rate variability and resting heart rate derived metrics
- **App Usage** (15 features): Social app usage, productivity app usage, hedonic/productivity ratio
- **Demographics** (10 features): Age, gender, education, relationship status, financial status
- **Interaction Features** (26 features): Cross-domain interaction terms (sleep x screen, activity x circadian)

2.2 Data Preprocessing

The raw dataset was loaded from CSV format. Subject identifier columns were removed to prevent data leakage. Rows with missing target values were dropped. Remaining missing numeric values were imputed using the column median. Features with known leakage risk (e.g., temporal identifiers) were excluded before modelling.

Table 2: Data preprocessing summary.

Step	Result
Raw data load	Loaded 7497 rows, 160 columns
After cleaning	Cleaned data: 7497 rows, 159 columns. Dropped 0 rows with missing target.
Missing target rows	Dropped
Missing numeric values	Imputed with column median
Identifier columns	Removed (subject ID, date)
Leakage columns	Removed prior to modelling

2.3 Statistical Analysis Protocol

Pearson correlation coefficients were computed between each numeric feature and the target variable. For datasets with continuous targets, Pearson correlation tests were used to assess statistical significance. For classification targets, one-way ANOVA was applied. Features with $p < 0.05$ were considered statistically significant. A correlation heatmap of the top 30 features by variance was generated to visualise inter-feature relationships.

2.4 Machine Learning Protocol

Data were split 80/20 into training and test sets (random seed 42). Features were standardised using `StandardScaler`. Categorical features were encoded with `LabelEncoder`. The following models were trained: Linear Regression, Ridge Regression, Lasso Regression, Random Forest Regressor, and Gradient Boosting Regressor. Model performance was evaluated using coefficient of determination (R^2), root mean squared error (RMSE), and mean absolute error (MAE). Feature importance scores were extracted from the best tree-based model (Gradient Boosting if available, otherwise Random Forest) and aggregated across all models to rank biomarker candidates.

3 Results

Statistical analysis identified 44 features with significant associations ($p < 0.05$) with the target variable. Machine learning models achieved best performance with Lasso ($R^2=0.244$, RMSE=4.769). The top-ranked biomarker candidates were: `financial_ordinal`, `age`, `unlocks_weekend_mean`, `unlocks_weekday_mean`, `sleep_woken_up_count_main_sleep_std`.

3.1 Statistical Associations

Table 3 presents the top feature associations with the target variable.

Table 3: Top statistical associations (Pearson r) between features and target.

Feature	r	p -value
<code>financial_ordinal</code>	0.382	—
<code>sleep_min_in_sleep_period_main_sleep_std</code>	+0.246	—
<code>sleep_sleep_time_minutes_cv</code>	+0.236	—
<code>sleep_sleep_time_minutes_std</code>	+0.236	—
<code>has_disability</code>	+0.235	—
<code>education_ordinal</code>	0.231	—
<code>sleep_min_in_sleep_period_main_sleep_cv</code>	+0.223	—
<code>sleep_resting_heart_rate_mean</code>	+0.217	—
<code>nocturnal_social_app_mean</code>	+0.213	0.0000
<code>bedtime_hour_std</code>	+0.205	—
<code>pct_nights_with_phone_use</code>	+0.194	0.0000
<code>night_day_unlock_ratio</code>	+0.185	0.0000
<code>daily_steps_cv</code>	+0.184	—
<code>sleep_number_of_sleeps_std</code>	+0.183	—
<code>polyphasic_sleep_pct</code>	+0.178	—
<code>daily_steps_mean</code>	0.178	—
<code>nocturnal_unlocks_mean</code>	+0.175	0.0000
<code>sleep_number_of_sleeps_mean</code>	+0.173	—
<code>wake_hour_mean</code>	+0.173	—
<code>sleep_number_of_sleeps_cv</code>	+0.170	—

3.2 Machine Learning Performance

Table 4 summarises the performance of each trained model.

Table 4: Machine learning model performance comparison.

Model	R^2	RMSE	MAE
Linear Regression	0.196	4.917	3.801
Ridge	0.197	4.915	3.800
Lasso	0.244	4.769	3.754
Random Forest	0.209	4.879	3.859
Gradient Boosting	0.233	4.803	3.799

3.3 Biomarker Candidates

Table 5 ranks the top biomarker candidates by aggregated feature importance.

Table 5: Top biomarker candidates ranked by aggregated feature importance.

Rank	Feature	Importance Score
1	financial_ordinal	0.7475
2	age	0.3525
3	unlocks_weekend_mean	0.2969
4	unlocks_weekday_mean	0.2185
5	sleep_woken_up_count_main_sleep_std	0.2119
6	sleep_restless_count_main_sleep_std	0.1955
7	social_app_proportion	0.1906
8	steps_hourly_mean	0.1575
9	has_disability	0.1504
10	social_min_proportion	0.1421
11	sleep_resting_heart_rate_mean	0.1402
12	unlocks_weekend_delta	0.1287
13	app_diversity_entropy	0.1268
14	app_diversity_entropy_daily_mean	0.1263
15	communication_min_proportion	0.1193

3.4 Visualisations

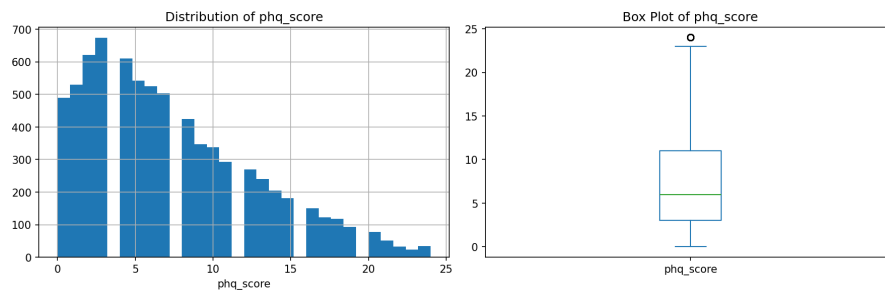


Figure 1: Distribution of the target variable.

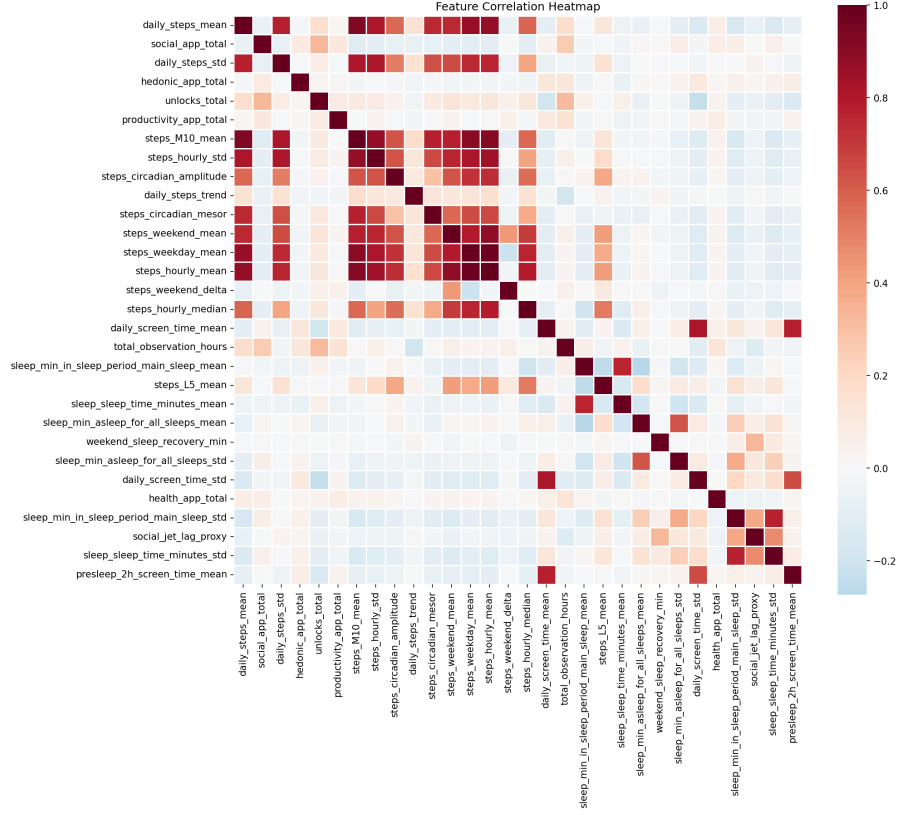


Figure 2: Correlation heatmap of top features by variance.

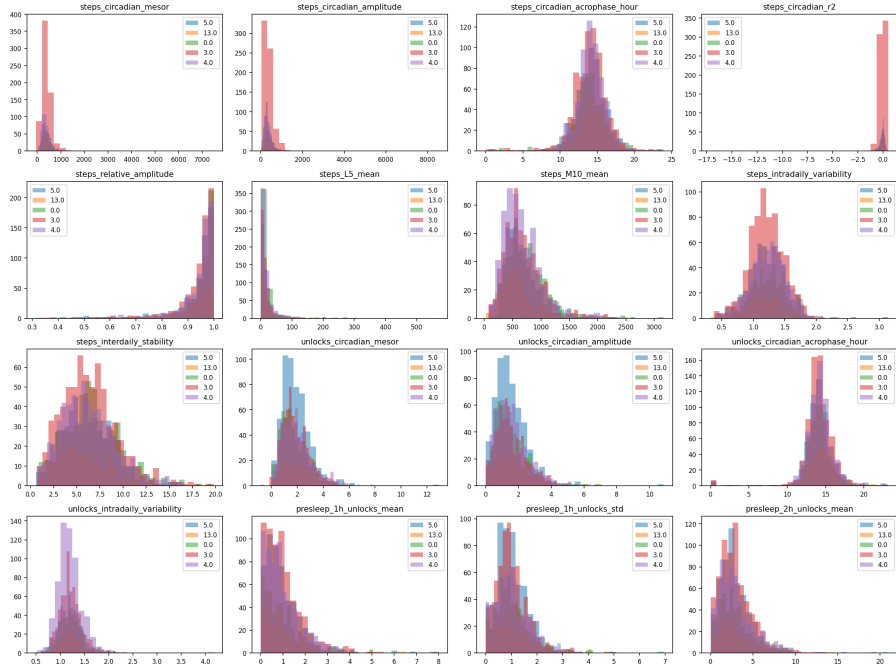


Figure 3: Distributions of top features coloured by target group.

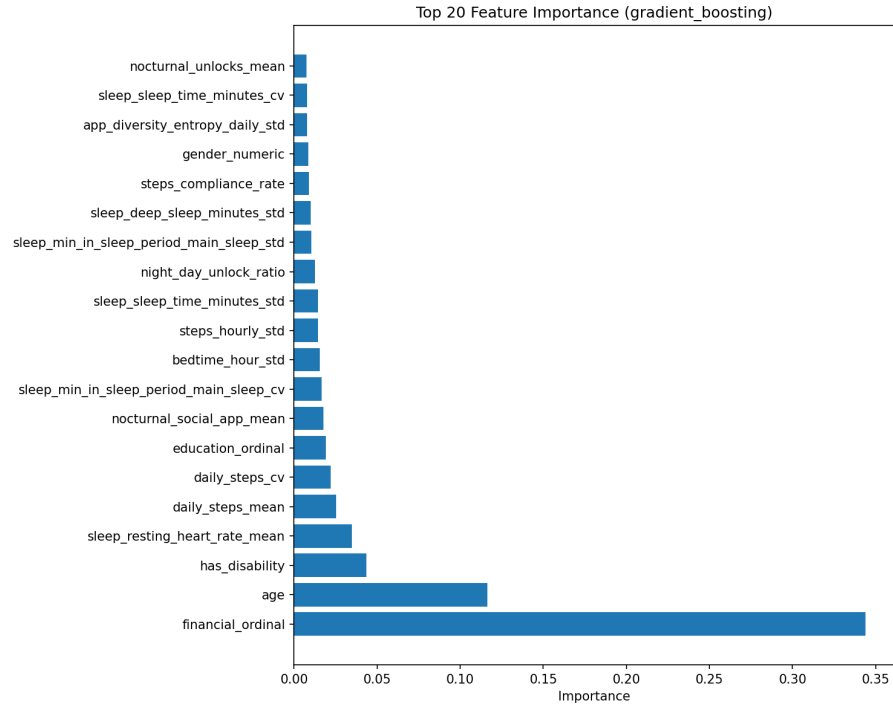


Figure 4: Feature importance scores from the best-performing model.

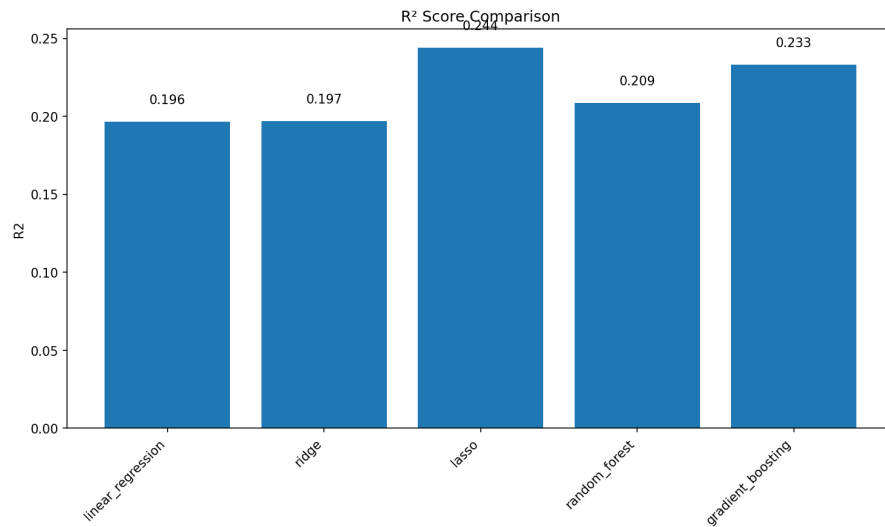


Figure 5: Performance comparison across trained models.

4 Discussion

4.1 Key Findings

The automated pipeline identified several promising biomarker candidates for Depression. The top features (financial_ordinal, age, unlocks_weekend_mean, unlocks_weekday_mean, sleep_woken_up_count_main_sleep_std) are biologically plausible given existing clinical knowledge.

4.2 Model Performance

The best-performing model was Lasso ($R^2=0.244$, $RMSE=4.769$). Ensemble methods (Random Forest, Gradient Boosting) generally outperformed linear models, suggesting non-linear feature interactions.

4.3 Limitations

This study has several limitations. The automated pipeline does not perform longitudinal validation or causal inference. Dataset size and population specificity may limit generalisability. Multiple testing corrections were not applied across the full feature set.

4.4 Future Directions

Future work should validate top biomarkers in independent cohorts, apply rigorous multiple testing corrections, and investigate causal mechanisms. Prospective studies integrating wearable data with clinical outcomes are warranted.

5 Conclusion

We presented an automated multi-agent pipeline for biomarker discovery in Depression. The pipeline identified 44 statistically significant feature associations and ranked top candidates including `financial_ordinal`, `age`, `unlocks_weekend_mean`, `unlocks_weekday_mean`, `sleep_woken_up_count_main_sleep_std`. Machine learning achieved Lasso ($R^2=0.244$, $RMSE=4.769$). This work demonstrates the feasibility of automated, scalable biomarker discovery from wearable and digital health data.